

INTERNATIONAL GCSE PHYSICS

9203/2

Paper 2

Mark scheme

November 2023

Version: 1.0 Final



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols/formulae

If a student writes a chemical symbol/formula instead of a required chemical name, full credit can be given if the symbol/formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(.....) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	protons	this order only	1	AO1 3.7.1f
	neutrons		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	nuclear fusion		1	AO1 3.7.4d 3.8.1b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	helium		1	AO1 3.8.1b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	temperature	allow pressure any order	1	AO1 3.8.1c
	density		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	millions of years		1	AO3 3.8.1b 3.8.1i

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	red giant		1	AO1 3.8.1j

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.7	black hole		1	AO1 3.8.1j
	neutron star		1	

Total Question 1		10
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Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	$a = \frac{30}{12}$		1	AO2 3.1.3d
	$a = 2.5 \text{ (m/s}^2\text{)}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	$p = 1500 \times 30$		1	AO2 3.1.4a
	$p = 45\,000 \text{ (kg m/s)}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	friction		1	AO2 3.1.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	any two from: - condition of the brakes - condition of the tyres - type of road surface - icy / wet conditions - mass of the lorry - gradient of the road		2	AO3 3.1.5e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	as the speed of the lorry increases the length of marks on the road increases	allow the relationship is non-linear	1	AO3 3.1.5e
	at a decreasing rate		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	20 (m/s)		1	AO3 3.1.5e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.7	$2\,380\,000 = F \times 34$		1	1 × AO1 3 × AO2 3.2.1a
	$F = \frac{2\,380\,000}{34}$		1	
	$F = 70\,000$		1	
	N		1	

Total Question 2		14
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	Level 3: The method would lead to the production of a valid outcome. The key steps are identified and logically sequenced.		5–6	AO4 3.4.1e
	Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the method is not fully logically sequenced.		3–4	
	Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.		1–2	
	No relevant content		0	
	Indicative content • put stearic acid into a test tube using the funnel • boil the water in the kettle • pour the boiling water into a beaker • place the test tube of stearic acid into the beaker • leave until the stearic acid melts • take the test tube out of the hot water • place the thermometer in the stearic acid • measure the temperature with the thermometer • start the stop clock • record the temperature every 30 seconds (or suitable time) • stir the stearic acid during experiment • continue to record the temperature after the stearic acid has all changed state			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	from A to B (particles are free to move but) particles are slowing down	allow kinetic energy is decreasing if no other mark awarded allow 1 mark for particles are free to move in a liquid but are vibrating about fixed position in a solid	1	1 × AO3 2 × AO2 3.4.1a
	from B to C particles change from being free to move to vibrating about fixed position		1	
	from C to D (vibrating about fixed position) but magnitude of vibration is decreasing		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	$3600 = 0.018 \times L_f$	allow correct rearrangement using an incorrectly / not converted value of m allow correct calculation using an incorrectly / not converted value of E	1	AO2 3.4.1c
	$L_f = \frac{3600}{0.018}$		1	
	$L_f = 200\,000 \text{ (J/kg)}$		1	

Total Question 3	12
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Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	kinetic energy (of the satellite) decreases		1	AO2 3.2.2a
	gravitational potential energy (of the satellite) increases		1	
	energy is dissipated to the surroundings	allow thermal energy of the satellite increases	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	gravitational force	allow gravity allow weight	1	1 × AO1 1 × AO2 3.1.1a
	air resistance	allow drag	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	both forces act in the opposite direction to motion		1	2 × AO3 1 × AO2 3.1.3h
	so there is a resultant force (in the opposite direction to the motion)	allow newton's second law	1	
	causes an acceleration in the opposite direction to the motion		1	
	OR both forces act in the opposite direction to motion (1) so do work on the satellite (1) which reduces the kinetic energy (of the satellite) (1)			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	$E_k = 625\,000\,000 \text{ (J)}$		1	AO2 3.2.1c
	$625\,000\,000 = \frac{1}{2} \times 200 \times v^2$	allow correct substitution of incorrectly / not converted value of E_k	1	
	$v^2 = \frac{2 \times 625\,000\,000}{200}$ or	allow correct rearrangement of incorrectly / not converted value of E_k	1	
	$v = \sqrt{\frac{2 \times 625\,000\,000}{200}}$ $v = 2500 \text{ (m/s)}$	allow correct calculation using incorrectly / not converted value of E_k	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	make the diameter of the drum greater		1	AO3 3.8.2g
	or			
	make the drum rotate faster so launch speed is greater		1	
	so height reached (above the surface of the Earth) is greater	dependent on MP1 or MP2 being awarded	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	D		1	AO2 3.8.2e 6-7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.7	acceleration is rate of change of velocity and velocity is a vector the direction of the velocity is changing (but the magnitude stays the same) or the Moon's direction is constantly changing (1) so there must be a resultant force (1) which causes the velocity to constantly change (1)	allow velocity has both magnitude and direction	1 1 1	2 × AO1 1 × AO2 3.8.2e

Total Question 4		19
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Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	reduces the current (for a given power)	allow makes the system more efficient	1	AO1 3.6.2b
	(so) less heating		1	
	(therefore) reducing the energy losses		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	to make it safer		1	AO2 3.6.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	$\frac{7200}{V_s} = 30$	allow correct rearrangement using incorrectly / not converted value of V_s	1	AO2 3.6.2e
	$V_s = \frac{7200}{30}$		1	
	$V_s = 240 \text{ (V)}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	advantage less likely to be damaged by high winds or falling trees or no visual pollution	allow (the street) looks better	1	AO3 3.6.2a
	disadvantage more expensive to install / repair or repairs are longer / difficult	do not allow more expensive on its own allow more difficult to identify (position of) fault	1	

Total Question 5		9
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Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	circuit with a cell, variable resistor and filament lamp in series		1	AO4 3.5.1g,m,n
	ammeter in series with the filament lamp and the cell		1	
	voltmeter in parallel with filament lamp		1	
		allow 2 marks if all symbols are correct but circuit is incorrect allow 2 marks if circuit is correct but 1 symbol incorrect allow 1 mark if at least 3 correct circuit symbols are seen		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	both axes correct with units		1	AO3 3.5.1m
	all 5 points plotted correctly		2	
	correct curved line of best fit through all the points		1	
		allow 1 mark if 3 or 4 points are correctly plotted		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	so that the cell / lamp would last longer	allow to reduce wasted energy	1	AO4 3.5.1m

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	(as temperature increases) amplitude of ion vibration increases		1	AO1 3.5.1m
	more frequent collisions between ion and electrons		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	$520 = \frac{E}{1.1}$		1	AO2 3.4.1b 3.6.5b 3.6.5c
	$E = 572 \text{ (J)}$		1	
	$572 = m \times 130 \times 2200$	for subsequent marks to be awarded the correct equation must have been used to calculate E	1	
	$m = \frac{572}{(130 \times 2200)}$		1	
	$m = 0.002 \text{ (kg)}$		1	
	$m = 2 \text{ (g)}$	allow 2.0 (g)	1	

Total Question 6		16
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Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	$\lambda = 6.56 \times 10^{-7} \text{ (m)}$		1	AO2 3.3.1h
	$3.00 \times 10^8 = f \times 6.56 \times 10^{-7}$		1	
	$f = \frac{3.00 \times 10^8}{6.56 \times 10^{-7}}$		1	
	or $4.57317... \times 10^{14}$			
	$f = 4.6 \times 10^{14} \text{ Hz}$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	the detected wavelength is longer		1	3 × AO3 3 × AO1 3.8.3b
	so light is red shifted		1	
	because galaxy is moving away from us		1	
	therefore the universe expanding		1	
	so must have been smaller in the past		1	
	therefore started from a single point		1	

Total Question 7		10
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